

## **Building of the Future: Intelligent Control of Mobile Structures**

### **Kurze Fassung:**

Mobile heating units consume enormous amounts of energy in tents, halls, and temporary accommodations, often simply because they are kept running continuously or are manually controlled.

With the possibility of remotely controlling heating devices, it is already possible to reduce operating costs. But that is only the beginning: with predictive control, smart fault detection, and compelling visualization of the savings, considerably more potential can be unlocked both financially and ecologically.

💡 A unique opportunity to build something that can directly impact the environment, shape an already existing product on the market, and offers the potential for a working student or internship position at LB Energy.

🧠 Skills: Everything is welcome

🔧 Support: if real data is needed it will be provided

### **Lange Fassung:**

How should mobile structures be controlled in the future to be maximally energy-efficient, environmentally conscious, and user-friendly? Mobile heating units consume enormous amounts of energy in tents, halls, and temporary accommodations, often simply because they are kept running continuously or controlled manually. With the Intelligent Heat Link (IHL) by LB Energy, operating costs have already been reduced by 20–30%. But that is only the beginning: with predictive control, smart fault detection, and compelling visualization of the savings, considerably more potential can be unlocked both financially and ecologically.

Picture the TUM Aerospace and Geodesy Campus in Ottobrunn-Taufkirchen. Eleven lecture halls are already in use, and the entire campus is climatized by 29 heat pumps. At 7 a.m., the first students arrive for an early lecture. By 10 a.m., another hall fills up, while two others stay empty until the afternoon.

When should each heat pump switch on, so that every hall reaches the right temperature exactly when the lecture starts and holds it until the end? What internal and external factors such as weather forecast, humidity, or the number of students within the hall are relevant to building a model that predicts the most energy efficient start time of the heat pump? What happens when conditions occur that the model has never seen before, such as an unusually cold spring morning or a sudden heatwave?

For such a system to work in real-world operation, it must also be reliable when something goes wrong. If a heater fails on a cold winter day, we should not have to learn about it from freezing students. This is where an early-warning system comes in. Based on sensor data, such as room temperature and others, it can reliably tell apart what is normal and what is not: short temperature drops when a door opens, gradual changes during the day, a real heater defect, or even a tampering attempt. As soon as something looks wrong, the system automatically alerts the building's superintendent or technician, so the problem gets fixed before anyone in the room notices.

For site managers or public authorities to understand the value of such a system and trust it, they need more than models running in the background. They need a visualization that clearly shows what intelligent control delivers in kWh saved, money saved, and CO<sub>2</sub> emissions avoided, broken down by event, by hall, by season. Other questions we continuously ask ourselves are: what happens if I lower the temperature by 2 °C? Would four units be enough instead of six? What other scenarios might be useful?

All these questions feed into the same vision: a mobile building that thinks for itself. Predictive algorithms need reliable anomaly detection so that operators can trust them. Reliable systems unfold their value only when their impact is made visible. We want to see how you bring these building blocks together, whether you dive deep into one aspect or combine several.